

2.B Classical Applications

In this section we use homology theory to prove several interesting results in topology and algebra whose statements give no hint that algebraic topology might be involved.

To begin, we calculate the homology of complements of embedded spheres and disks in a sphere. Recall that an embedding is a map that is a homeomorphism onto its image.

Proposition 2B.1. (a) *For an embedding $h: D^k \rightarrow S^n$, $\tilde{H}_i(S^n - h(D^k)) = 0$ for all i .*
 (b) *For an embedding $h: S^k \rightarrow S^n$ with $k < n$, $\tilde{H}_i(S^n - h(S^k))$ is $\mathbb{Z}$ for $i = n - k - 1$ and 0 otherwise.*

As a special case of (b) we have the Jordan curve theorem: A subspace of S^2 homeomorphic to S^1 separates S^2 into two complementary components, or equivalently, path-components since open subsets of S^n are locally path-connected. One could just as well use $\mathbb{R}^2$ in place of S^2 here since deleting a point from an open set in S^2 does not affect its connectedness. More generally, (b) says that a subspace of S^n homeomorphic to S^{n-1} separates it into two components, and these components have the same homology groups as a point. Somewhat surprisingly, there are embeddings where these complementary components are not simply-connected as they are for the standard embedding. An example is the Alexander horned sphere in S^3 which we describe in detail following the proof of the proposition. These complications involving embedded S^{n-1} 's in S^n are all local in nature since it is known that any locally nicely embedded S^{n-1} in S^n is equivalent to the standard $S^{n-1} \subset S^n$, equivalent in the sense that there is a homeomorphism of S^n taking the given embedded S^{n-1} onto the standard S^{n-1} . In particular, both complementary regions are homeomorphic to open balls. See [Brown 1960] for a precise statement and proof. When $n = 2$ it is a classical theorem of Schoenflies that all embeddings $S^1 \hookrightarrow S^2$ are equivalent.

By contrast, when we come to embeddings of S^{n-2} in S^n , even locally nice embeddings need not be equivalent to the standard one. This is the subject of knot theory, including the classical case of knotted embeddings of S^1 in S^3 or $\mathbb{R}^3$. For embeddings of S^{n-2} in S^n the complement always has the same homology as S^1 , according to the theorem, but the fundamental group can be quite different. In spite of the fact that the homology of a knot complement does not detect knottedness, it is still possible to use homology to distinguish different knots by looking at the homology of covering spaces of their complements.

Proof: We prove (a) by induction on k . When $k = 0$, $S^n - h(D^0)$ is homeomorphic to $\mathbb{R}^n$, so this case is trivial. For the induction step it will be convenient to replace the domain disk D^k of h by the cube I^k . Let $A = S^n - h(I^{k-1} \times [0, 1/2])$ and let

$B = S^n - h(I^{k-1} \times [1/2, 1])$, so $A \cap B = S^n - h(I^k)$ and $A \cup B = S^n - h(I^{k-1} \times \{1/2\})$. By induction $\tilde{H}_i(A \cup B) = 0$ for all i , so the Mayer-Vietoris sequence gives isomorphisms $\Phi: \tilde{H}_i(S^n - h(I^k)) \rightarrow \tilde{H}_i(A) \oplus \tilde{H}_i(B)$ for all i . Modulo signs, the two components of Φ are induced by the inclusions $S^n - h(I^k) \hookrightarrow A$ and $S^n - h(I^k) \hookrightarrow B$, so if there exists an i -dimensional cycle α in $S^n - h(I^k)$ that is not a boundary in $S^n - h(I^k)$, then α is also not a boundary in at least one of A and B . (When $i = 0$ the word ‘cycle’ here is to be interpreted in the sense of augmented chain complexes since we are dealing with reduced homology.) By iteration we can then produce a nested sequence of closed intervals $I_1 \supset I_2 \supset \dots$ in the last coordinate of I^k shrinking down to a point $p \in I$, such that α is not a boundary in $S^n - h(I^{k-1} \times I_m)$ for any m . On the other hand, by induction on k we know that α is the boundary of a chain β in $S^n - h(I^{k-1} \times \{p\})$. This β is a finite linear combination of singular simplices with compact image in $S^n - h(I^{k-1} \times \{p\})$. The union of these images is covered by the nested sequence of open sets $S^n - h(I^{k-1} \times I_m)$, so by compactness β must actually be a chain in $S^n - h(I^{k-1} \times I_m)$ for some m . This contradiction shows that α must be a boundary in $S^n - h(I^k)$, finishing the induction step.

Part (b) is also proved by induction on k , starting with the trivial case $k = 0$ when $S^n - h(S^0)$ is homeomorphic to $S^{n-1} \times \mathbb{R}$. For the induction step, write S^k as the union of hemispheres D_+^k and D_-^k intersecting in S^{k-1} . The Mayer-Vietoris sequence for $A = S^n - h(D_+^k)$ and $B = S^n - h(D_-^k)$, both of which have trivial reduced homology by part (a), then gives isomorphisms $\tilde{H}_i(S^n - h(D^k)) \approx \tilde{H}_{i+1}(S^n - h(S^{k-1}))$. $\square$

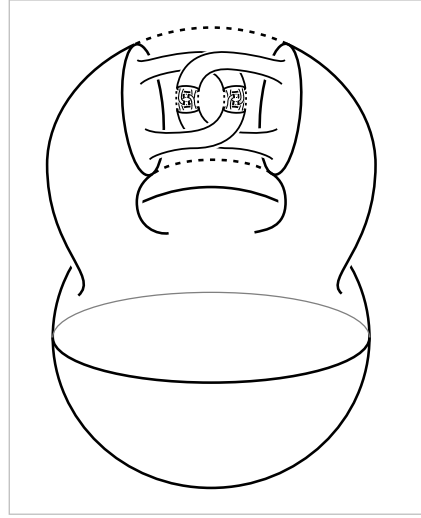
If we apply the last part of this proof to an embedding $h: S^n \rightarrow S^n$, the Mayer-Vietoris sequence ends with the terms $\tilde{H}_0(A) \oplus \tilde{H}_0(B) \rightarrow \tilde{H}_0(S^n - h(S^{n-1})) \rightarrow 0$. Both $\tilde{H}_0(A)$ and $\tilde{H}_0(B)$ are zero, so exactness would imply that $\tilde{H}_0(S^n - h(S^{n-1})) = 0$ which appears to contradict the fact that $S^n - h(S^{n-1})$ has two path-components. The only way out of this dilemma is for h to be surjective, so that $A \cap B$ is empty and the 0 at the end of the Mayer-Vietoris sequence is $\tilde{H}_{-1}(\emptyset)$ which is $\mathbb{Z}$ rather than 0.

In particular, this shows that S^n cannot be embedded in $\mathbb{R}^n$ since this would yield a nonsurjective embedding in S^n . A consequence is that there is no embedding $\mathbb{R}^m \hookrightarrow \mathbb{R}^n$ for $m > n$ since this would restrict to an embedding of $S^n \subset \mathbb{R}^m$ into $\mathbb{R}^n$. More generally there is no continuous injection $\mathbb{R}^m \rightarrow \mathbb{R}^n$ for $m > n$ since this too would give an embedding $S^n \hookrightarrow \mathbb{R}^n$.

Example 2B.2: The Alexander Horned Sphere. This is a subspace $S \subset \mathbb{R}^3$ homeomorphic to S^2 such that the unbounded component of $\mathbb{R}^3 - S$ is not simply-connected as it is for the standard $S^2 \subset \mathbb{R}^3$. We will construct S by defining a sequence of compact subspaces $X_0 \supset X_1 \supset \dots$ of $\mathbb{R}^3$ whose intersection is homeomorphic to a ball, and then S will be the boundary sphere of this ball.

We begin with X_0 a solid torus $S^1 \times D^2$ obtained from a ball B_0 by attaching a handle $I \times D^2$ along $\partial I \times D^2$. In the figure this handle is shown as the union of

two ‘horns’ attached to the ball, together with a shorter handle drawn as dashed lines. To form the space $X_1 \subset X_0$ we delete part of the short handle, so that what remains is a pair of linked handles attached to the ball B_1 that is the union of B_0 with the two horns. To form X_2 the process is repeated: Decompose each of the second stage handles as a pair of horns and a short handle, then delete a part of the short handle. In the same way X_n is constructed inductively from X_{n-1} . Thus X_n is a ball B_n with 2^n handles attached, and B_n is obtained from B_{n-1} by attaching 2^n horns. There are homeomorphisms $h_n: B_{n-1} \rightarrow B_n$ that are the identity



outside a small neighborhood of $B_n - B_{n-1}$. As n goes to infinity, the composition $h_n \cdots h_1$ approaches a map $f: B_0 \rightarrow \mathbb{R}^3$ which is continuous since the convergence is uniform. The set of points in B_0 where f is not equal to $h_n \cdots h_1$ for large n is a Cantor set, whose image under f is the intersection of all the handles. It is not hard to see that f is one-to-one. By compactness it follows that f is a homeomorphism onto its image, a ball $B \subset \mathbb{R}^3$ whose boundary sphere $f(\partial B_0)$ is S , the Alexander horned sphere.

Now we compute $\pi_1(\mathbb{R}^3 - B)$. Note that B is the intersection of the X_n 's, so $\mathbb{R}^3 - B$ is the union of the complements Y_n of the X_n 's, which form an increasing sequence $Y_0 \subset Y_1 \subset \cdots$. We will show that the groups $\pi_1(Y_n)$ also form an increasing sequence of successively larger groups, whose union is $\pi_1(\mathbb{R}^3 - B)$. To begin we have $\pi_1(Y_0) \approx \mathbb{Z}$ since X_0 is a solid torus embedded in $\mathbb{R}^3$ in a standard way. To compute $\pi_1(Y_1)$, let $\bar{Y}_0$ be the closure of Y_0 in Y_1 , so $\bar{Y}_0 - Y_0$ is an open annulus A and $\pi_1(\bar{Y}_0)$ is also $\mathbb{Z}$. We obtain Y_1 from $\bar{Y}_0$ by attaching the space $Z = Y_1 - Y_0$ along A . The group $\pi_1(Z)$ is the free group F_2 on two generators α_1 and α_2 represented by loops linking the two handles, since $Z - A$ is homeomorphic to an open ball with two straight tubes deleted. A loop α generating $\pi_1(A)$ represents the commutator $[\alpha_1, \alpha_2]$, as one can see by noting that the closure of Z is obtained from Z by adjoining two disjoint surfaces, each homeomorphic to a torus with an open disk removed; the boundary of this disk is homotopic to α and is also homotopic to the commutator of meridian and longitude circles in the torus, which correspond to α_1 and α_2 . Van Kampen's theorem now implies that the inclusion $Y_0 \hookrightarrow Y_1$ induces an injection of $\pi_1(Y_0)$ into $\pi_1(Y_1)$ as the infinite cyclic subgroup generated by $[\alpha_1, \alpha_2]$.

In a similar way we can regard Y_{n+1} as being obtained from Y_n by adjoining 2^n copies of Z . Assuming inductively that $\pi_1(Y_n)$ is the free group F_{2^n} with generators represented by loops linking the 2^n smallest handles of X_n , then each copy of Z ad-

joined to Y_n changes $\pi_1(Y_n)$ by making one of the generators into the commutator of two new generators. Note that adjoining a copy of Z induces an injection on π_1 since the induced homomorphism is the free product of the injection $\pi_1(A) \rightarrow \pi_1(Z)$ with the identity map on the complementary free factor. Thus the map $\pi_1(Y_n) \rightarrow \pi_1(Y_{n+1})$ is an injection $F_{2n} \rightarrow F_{2n+1}$. The group $\pi_1(\mathbb{R}^3 - B)$ is isomorphic to the union of this increasing sequence of groups by a compactness argument: Each loop in $\mathbb{R}^3 - B$ has compact image and hence must lie in some Y_n , and similarly for homotopies of loops.

In particular we see explicitly why $\pi_1(\mathbb{R}^3 - B)$ has trivial abelianization, because each of its generators is exactly equal to the commutator of two other generators. This inductive construction in which each generator of a free group is decreed to be the commutator of two new generators is perhaps the simplest way of building a nontrivial group with trivial abelianization, and for the construction to have such a nice geometric interpretation is something to marvel at. From a naive viewpoint it may seem a little odd that a highly nonfree group can be built as a union of an increasing sequence of free groups, but this can also easily happen for abelian groups, as $\mathbb{Q}$ for example is the union of an increasing sequence of infinite cyclic subgroups.

The next theorem says that for subspaces of $\mathbb{R}^n$, the property of being open is a topological invariant. This result is known classically as Invariance of Domain, the word ‘domain’ being an older designation for an open set in $\mathbb{R}^n$.

Theorem 2B.3. *If U is an open set in $\mathbb{R}^n$ then for any embedding $h: U \rightarrow \mathbb{R}^n$ the image $h(U)$ must be an open set in $\mathbb{R}^n$.*

Proof: Regarding S^n as the one-point compactification of $\mathbb{R}^n$, an equivalent statement is that $h(U)$ is open in S^n , and this is what we will prove. Each $x \in U$ is the center point of a disk $D^n \subset U$. It will suffice to prove that $h(D^n - \partial D^n)$ is open in S^n . By the previous proposition $S^n - h(\partial D^n)$ has two path-components. These path-components are $h(D^n - \partial D^n)$ and $S^n - h(D^n)$ since these two subspaces are disjoint and the first is path-connected since it is homeomorphic to $D^n - \partial D^n$ while the second is path-connected by the proposition. Since $S^n - h(\partial D^n)$ is open in S^n , its path-components are the same as its components. The components of a space with finitely many components are open, so $h(D^n - \partial D^n)$ is open in $S^n - h(\partial D^n)$ and hence also in S^n . $\square$

Here is an application involving the notion of an n -manifold, which is a Hausdorff space locally homeomorphic to $\mathbb{R}^n$:

Corollary 2B.4. *If M is a compact n -manifold and N is a connected n -manifold, then an embedding $h: M \rightarrow N$ must be surjective, hence a homeomorphism.*

Proof: $h(M)$ is closed in N since it is compact and N is Hausdorff. Since N is connected it suffices to show $h(M)$ is also open in N , and this is immediate from the theorem. $\square$

The Invariance of Domain and the n -dimensional generalization of the Jordan curve theorem were first proved by Brouwer around 1910, at a very early stage in the development of algebraic topology.

Division Algebras

Here is an algebraic application of homology theory due to H. Hopf:

Theorem 2B.5. *$\mathbb{R}$ and $\mathbb{C}$ are the only finite-dimensional division algebras over $\mathbb{R}$ which are commutative and have an identity.*

By definition, an **algebra** structure on $\mathbb{R}^n$ is simply a bilinear multiplication map $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, $(a, b) \mapsto ab$. Thus the product satisfies left and right distributivity, $a(b+c) = ab+ac$ and $(a+b)c = ac+bc$, and scalar associativity, $\alpha(ab) = (\alpha a)b = a(\alpha b)$ for $\alpha \in \mathbb{R}$. Commutativity, full associativity, and an identity element are not assumed. An algebra is a **division algebra** if the equations $ax = b$ and $xa = b$ are always solvable whenever $a \neq 0$. In other words, the linear transformations $x \mapsto ax$ and $x \mapsto xa$ are surjective when $a \neq 0$. These are linear maps $\mathbb{R}^n \rightarrow \mathbb{R}^n$, so surjectivity is equivalent to having trivial kernel, which means there are no zero-divisors.

The four classical examples are $\mathbb{R}$, $\mathbb{C}$, the quaternions $\mathbb{H}$, and the octonions $\mathbb{O}$. Frobenius proved in 1877 that $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$ are the only finite-dimensional associative division algebras over $\mathbb{R}$, and in 1898 Hurwitz proved that these three together with $\mathbb{O}$ are the only finite-dimensional division algebras over $\mathbb{R}$ with a product satisfying $|ab| = |a||b|$. See [Ebbinghaus 1991]. We will show in Theorem 3.20 that a finite-dimensional division algebra over $\mathbb{R}$ must have dimension a power of 2. In fact the only possible dimensions are 1, 2, 4, and 8, as in the classical examples. The first proofs of this appeared in [Bott & Milnor 1958] and [Kervaire 1958]. A very nice proof using K-theory is in [Adams & Atiyah 1966], and an exposition of this can be found in [VBKT]. See §4.B for further comments. It still appears that the only known proofs of this seemingly algebraic result are topological.

Proof: Suppose first that $\mathbb{R}^n$ has a commutative division algebra structure. Define a map $f: S^{n-1} \rightarrow S^{n-1}$ by $f(x) = x^2/|x^2|$. This is well-defined since $x \neq 0$ implies $x^2 \neq 0$ in a division algebra. The map f is continuous since the multiplication map $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is bilinear, hence continuous. Since $f(-x) = f(x)$ for all x , f induces a quotient map $\bar{f}: \mathbb{R}P^{n-1} \rightarrow S^{n-1}$. The following argument shows that $\bar{f}$ is injective. An equality $f(x) = f(y)$ implies $x^2 = \alpha^2 y^2$ for $\alpha = (|x^2|/|y^2|)^{1/2} > 0$. Thus we have $x^2 - \alpha^2 y^2 = 0$, which factors as $(x + \alpha y)(x - \alpha y) = 0$ using commutativity and the fact that α is a real scalar. Since there are no divisors of zero, we deduce that $x = \pm \alpha y$. Since x and y are unit vectors and α is real, this yields $x = \pm y$, so x and y determine the same point of $\mathbb{R}P^{n-1}$, which means that $\bar{f}$ is injective.

Since $\bar{f}$ is an injective map of compact Hausdorff spaces, it must be a homeomorphism onto its image. By Corollary 2B.4, $\bar{f}$ must in fact be surjective if we are

not in the trivial case $n = 1$. Thus we have a homeomorphism $\mathbb{R}P^{n-1} \approx S^{n-1}$. This implies $n = 2$ since if $n > 2$ the spaces $\mathbb{R}P^{n-1}$ and S^{n-1} have different homology groups (or different fundamental groups).

It remains to show that a 2-dimensional commutative division algebra A with identity is isomorphic to $\mathbb{C}$. This is elementary algebra: If $j \in A$ is not a real scalar multiple of the identity element $1 \in A$ and we write $j^2 = a + bj$ for $a, b \in \mathbb{R}$, then $(j - b/2)^2 = a + b^2/4$ so by rechoosing j we may assume that $j^2 = a \in \mathbb{R}$. If $a \geq 0$, say $a = c^2$, then $j^2 = c^2$ implies $(j + c)(j - c) = 0$, so $j = \pm c$, but this contradicts the choice of j . So $j^2 = -c^2$ and by rescaling j we may assume $j^2 = -1$, hence A is isomorphic to $\mathbb{C}$. $\square$

Leaving out the last paragraph, the proof shows that a finite-dimensional commutative division algebra, not necessarily with an identity, must have dimension at most 2. Oddly enough, there do exist 2-dimensional commutative division algebras without identity elements, for example $\mathbb{C}$ with the modified multiplication $z \cdot w = \overline{z}w$, the bar denoting complex conjugation.

The Borsuk-Ulam Theorem

In Theorem 1.10 we proved the 2-dimensional case of the Borsuk-Ulam theorem, and now we will give a proof for all dimensions, using the following theorem of Borsuk:

Proposition 2B.6. *An odd map $f: S^n \rightarrow S^n$, satisfying $f(-x) = -f(x)$ for all x , must have odd degree.*

The corresponding result that even maps have even degree is easier, and was an exercise for §2.2.

The proof will show that using homology with a coefficient group other than $\mathbb{Z}$ can sometimes be a distinct advantage. The main ingredient will be a certain exact sequence associated to a two-sheeted covering space $p: \tilde{X} \rightarrow X$,

$$\cdots \rightarrow H_n(X; \mathbb{Z}_2) \xrightarrow{\tau_*} H_n(\tilde{X}; \mathbb{Z}_2) \xrightarrow{p_*} H_n(X; \mathbb{Z}_2) \rightarrow H_{n-1}(X; \mathbb{Z}_2) \rightarrow \cdots$$

This is the long exact sequence of homology groups associated to a short exact sequence of chain complexes consisting of short exact sequences of chain groups

$$0 \rightarrow C_n(X; \mathbb{Z}_2) \xrightarrow{\tau} C_n(\tilde{X}; \mathbb{Z}_2) \xrightarrow{p_\#} C_n(X; \mathbb{Z}_2) \rightarrow 0$$

The map $p_\#$ is surjective since singular simplices $\sigma: \Delta^n \rightarrow X$ always lift to $\tilde{X}$, as Δ^n is simply-connected. Each σ has in fact precisely two lifts $\tilde{\sigma}_1$ and $\tilde{\sigma}_2$. Because we are using $\mathbb{Z}_2$ coefficients, the kernel of $p_\#$ is generated by the sums $\tilde{\sigma}_1 + \tilde{\sigma}_2$. So if we define τ to send each $\sigma: \Delta^n \rightarrow X$ to the sum of its two lifts to $\tilde{X}$, then the image of τ is the kernel of $p_\#$. Obviously τ is injective, so we have the short exact sequence indicated. Since τ and $p_\#$ commute with boundary maps, we have a short exact sequence of chain complexes, yielding the long exact sequence of homology groups.

The map τ_* is a special case of more general *transfer homomorphisms* considered in §3.G, so we will refer to the long exact sequence involving the maps τ_* as the *transfer sequence*. This sequence can also be viewed as a special case of the Gysin sequences discussed in §4.D. There is a generalization of the transfer sequence to homology with other coefficients, but this uses a more elaborate form of homology called homology with local coefficients, as we show in §3.H.

Proof of 2B.6: The proof will involve the transfer sequence for the covering space $p: S^n \rightarrow \mathbb{R}P^n$. This has the following form, where to simplify notation we abbreviate $\mathbb{R}P^n$ to P^n and we let the coefficient group $\mathbb{Z}_2$ be implicit:

$$\begin{aligned} 0 \longrightarrow H_n(P^n) &\xrightarrow[\approx]{\tau_*} H_n(S^n) \xrightarrow[0]{p_*} H_n(P^n) \xrightarrow[\approx]{} H_{n-1}(P^n) \longrightarrow 0 \longrightarrow \cdots \\ &\cdots \longrightarrow 0 \longrightarrow H_i(P^n) \xrightarrow[\approx]{} H_{i-1}(P^n) \longrightarrow 0 \longrightarrow \cdots \\ &\cdots \longrightarrow 0 \longrightarrow H_1(P^n) \xrightarrow[\approx]{} H_0(P^n) \xrightarrow[0]{} H_0(S^n) \xrightarrow[\approx]{p_*} H_0(P^n) \longrightarrow 0 \end{aligned}$$

The initial 0 is $H_{n+1}(P^n; \mathbb{Z}_2)$, which vanishes since P^n is an n -dimensional CW complex. The other terms that are zero are $H_i(S^n)$ for $0 < i < n$. We assume $n > 1$, leaving the minor modifications needed for the case $n = 1$ to the reader. All the terms that are not zero are $\mathbb{Z}_2$, by cellular homology. Alternatively, this exact sequence can be used to compute the homology groups $H_i(\mathbb{R}P^n; \mathbb{Z}_2)$ if one does not already know them. Since all the nonzero groups in the sequence are $\mathbb{Z}_2$, exactness forces the maps to be isomorphisms or zero as indicated.

An odd map $f: S^n \rightarrow S^n$ induces a quotient map $\bar{f}: \mathbb{R}P^n \rightarrow \mathbb{R}P^n$. These two maps induce a map from the transfer sequence to itself, and we will need to know that the squares in the resulting diagram commute. This follows from the naturality of the long exact sequence of homology associated to a short exact sequence of chain complexes, once we verify commutativity of the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & C_i(P^n) & \xrightarrow{\tau} & C_i(S^n) & \xrightarrow{p_\#} & C_i(P^n) \longrightarrow 0 \\ & & \downarrow \bar{f}_\# & & \downarrow f_\# & & \downarrow \bar{f}_\# \\ 0 & \longrightarrow & C_i(P^n) & \xrightarrow{\tau} & C_i(S^n) & \xrightarrow{p_\#} & C_i(P^n) \longrightarrow 0 \end{array}$$

Here the right-hand square commutes since $p\bar{f} = \bar{f}p$. The left-hand square commutes since for a singular i -simplex $\sigma: \Delta^i \rightarrow P^n$ with lifts $\tilde{\sigma}_1$ and $\tilde{\sigma}_2$, the two lifts of $\bar{f}\sigma$ are $f\tilde{\sigma}_1$ and $f\tilde{\sigma}_2$ since f takes antipodal points to antipodal points.

Now we can see that all the maps f_* and $\bar{f}_*$ in the commutative diagram of transfer sequences are isomorphisms by induction on dimension, using the evident fact that if three maps in a commutative square are isomorphisms, so is the fourth. The induction starts with the trivial fact that f_* and $\bar{f}_*$ are isomorphisms in dimension zero.

In particular we deduce that the map $f_*: H_n(S^n; \mathbb{Z}_2) \rightarrow H_n(S^n; \mathbb{Z}_2)$ is an isomorphism. By Lemma 2.49 this map is multiplication by the degree of $f \bmod 2$, so the degree of f must be odd. $\square$

The fact that odd maps have odd degree easily implies the Borsuk-Ulam theorem:

Corollary 2B.7. *For every map $g: S^n \rightarrow \mathbb{R}^n$ there exists a point $x \in S^n$ with $g(x) = g(-x)$.*

Proof: Let $f(x) = g(x) - g(-x)$, so f is odd. We need to show that $f(x) = 0$ for some x . If this is not the case, we can replace $f(x)$ by $f(x)/|f(x)|$ to get a new map $f: S^n \rightarrow S^{n-1}$ which is still odd. The restriction of this f to the equator S^{n-1} then has odd degree by the proposition. But this restriction is nullhomotopic via the restriction of f to one of the hemispheres bounded by S^{n-1} . $\square$

Exercises

1. Compute $H_i(S^n - X)$ when X is a subspace of S^n homeomorphic to $S^k \vee S^\ell$ or to $S^k \amalg S^\ell$.
2. Show that $\tilde{H}_i(S^n - X) \approx \tilde{H}_{n-i-1}(X)$ when X is homeomorphic to a finite connected graph. [First do the case that the graph is a tree.]
3. Let $(D, S) \subset (D^n, S^{n-1})$ be a pair of subspaces homeomorphic to (D^k, S^{k-1}) , with $D \cap S^{n-1} = S$. Show the inclusion $S^{n-1} - S \hookrightarrow D^n - D$ induces an isomorphism on homology. [Glue two copies of (D^n, D) together along (S^{n-1}, S) and examine the Mayer-Vietoris sequence for the complement of the resulting k -sphere in S^n , decomposed into two copies of $D^n - D$.]
4. In the unit sphere $S^{p+q-1} \subset \mathbb{R}^{p+q}$ let S^{p-1} and S^{q-1} be the subspheres consisting of points whose last q and first p coordinates are zero, respectively.
 - (a) Show that $S^{p+q-1} - S^{p-1}$ deformation retracts onto S^{q-1} , and is in fact homeomorphic to $S^{q-1} \times \mathbb{R}^p$.
 - (b) Show that S^{p-1} and S^{q-1} are not the boundaries of any pair of disjointly embedded disks D^p and D^q in D^{p+q} . [The preceding exercise may be useful.]
5. Let S be an embedded k -sphere in S^n for which there exists a disk $D^n \subset S^n$ intersecting S in the disk $D^k \subset D^n$ defined by the first k coordinates of D^n . Let $D^{n-k} \subset D^n$ be the disk defined by the last $n-k$ coordinates, with boundary sphere S^{n-k-1} . Show that the inclusion $S^{n-k-1} \hookrightarrow S^n - S$ induces an isomorphism on homology groups.
6. Modify the construction of the Alexander horned sphere to produce an embedding $S^2 \hookrightarrow \mathbb{R}^3$ for which neither component of $\mathbb{R}^3 - S^2$ is simply-connected.